

Image filtering

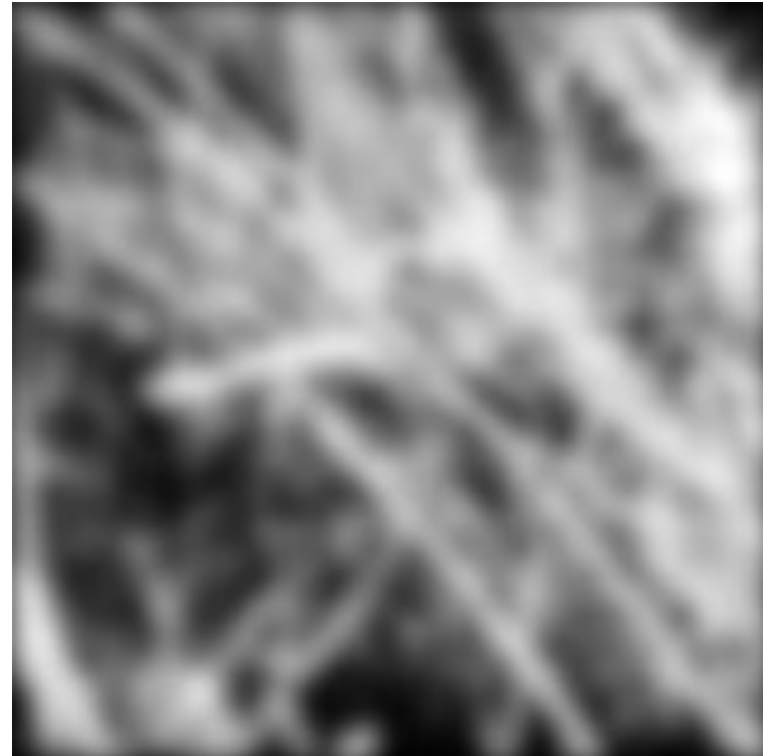


Image filtering: Outline

- Linear filtering and its properties
- Gaussian filters and their properties
- Nonlinear filtering: Median filtering, bilateral filtering
- Fun filtering application: Hybrid images

Sliding window operations

- Let's slide a fixed-size window over the image and perform the same simple computation at each window location
- Example use case: how do we reduce image noise?
 - Let's take the *average* of pixel values in each window
 - More generally, we can take a *weighted sum* where the weights are given by a *filter kernel*



Applying a linear filter

Input Filter Output

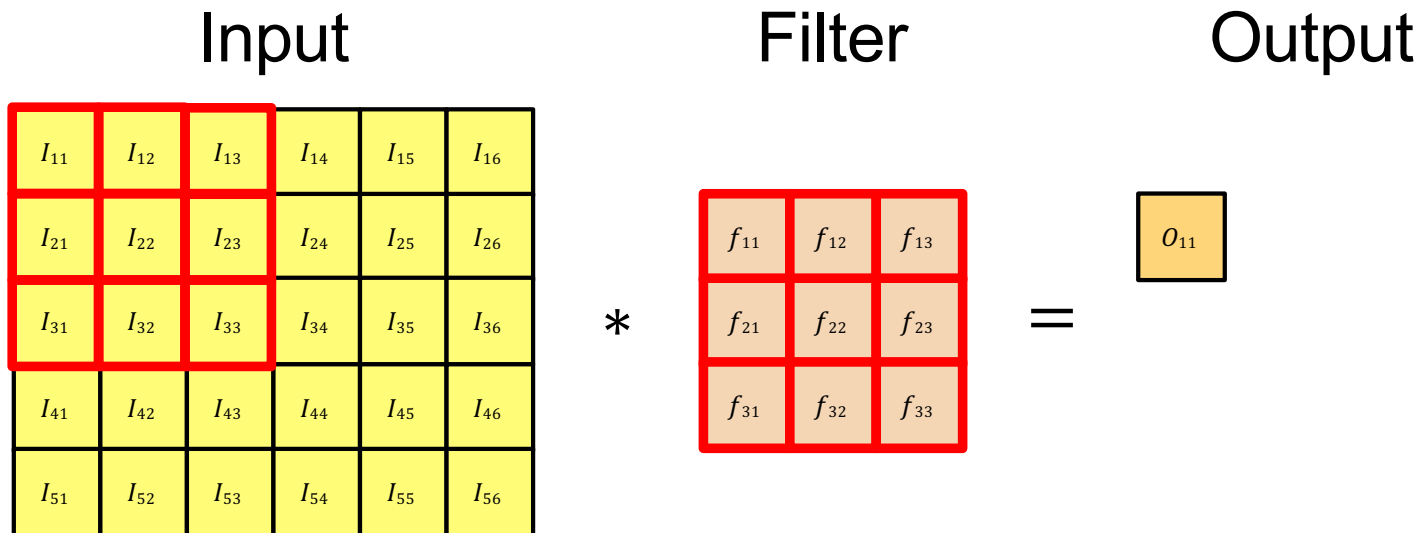
I_{11}	I_{12}	I_{13}	I_{14}	I_{15}	I_{16}
I_{21}	I_{22}	I_{23}	I_{24}	I_{25}	I_{26}
I_{31}	I_{32}	I_{33}	I_{34}	I_{35}	I_{36}
I_{41}	I_{42}	I_{43}	I_{44}	I_{45}	I_{46}
I_{51}	I_{52}	I_{53}	I_{54}	I_{55}	I_{56}

*

f_{11}	f_{12}	f_{13}
f_{21}	f_{22}	f_{23}
f_{31}	f_{32}	f_{33}

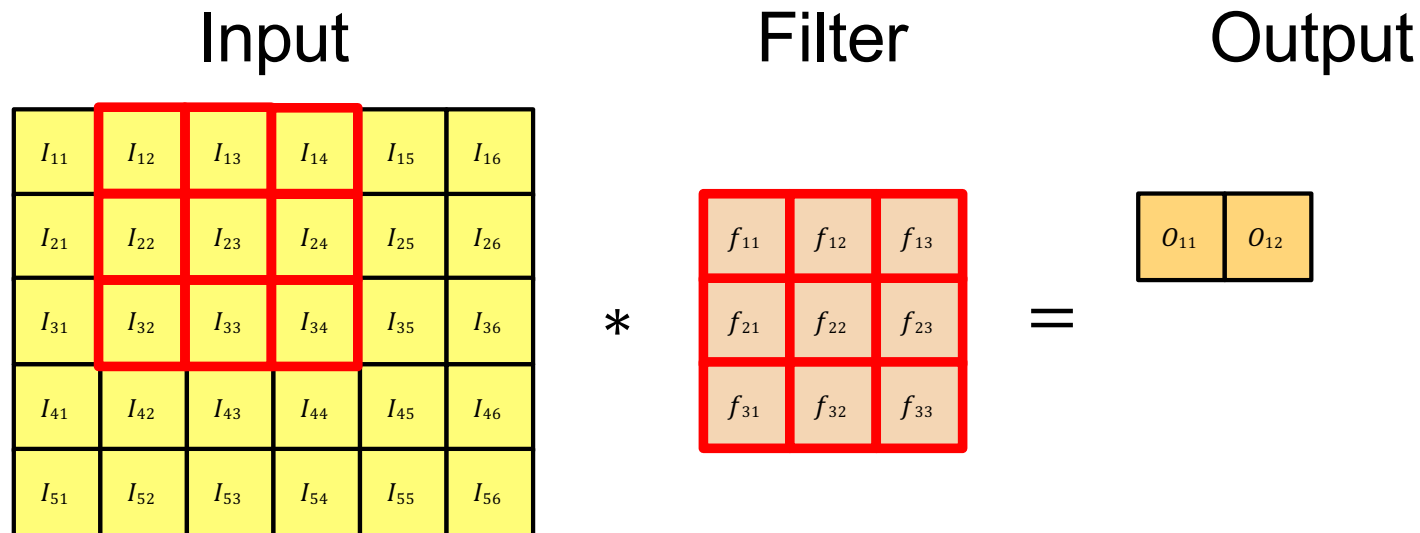
=

Applying a linear filter



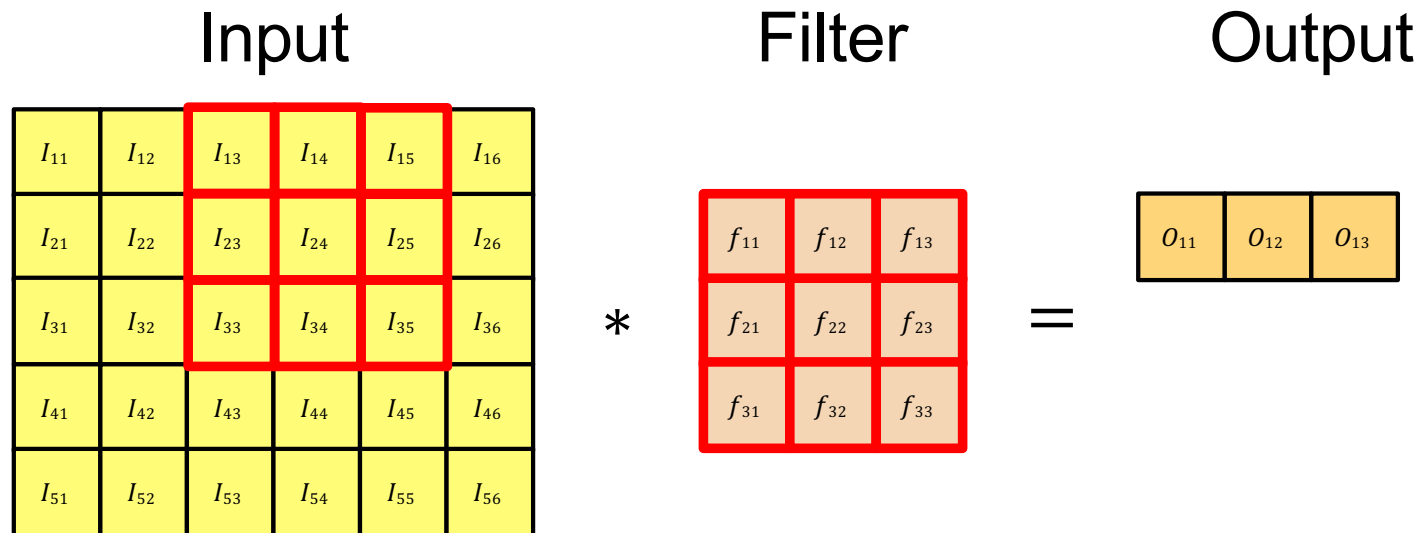
$$O_{11} = I_{11} \cdot f_{11} + I_{12} \cdot f_{12} + I_{13} \cdot f_{13} + \dots + I_{33} \cdot f_{33}$$

Applying a linear filter



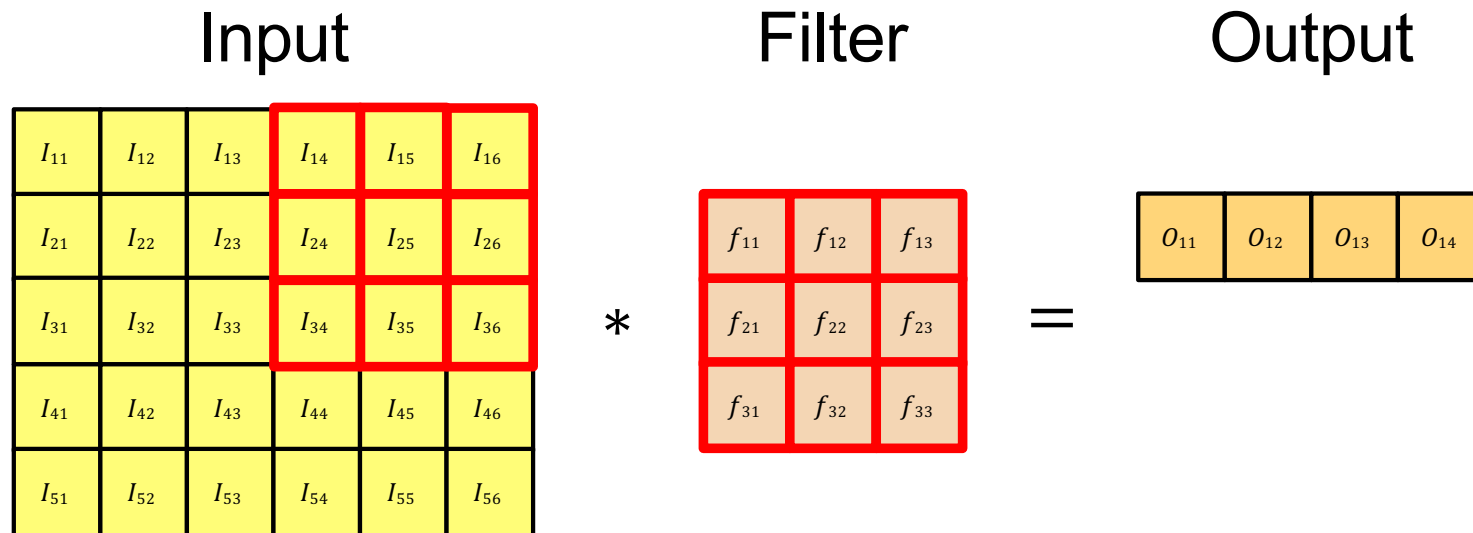
$$O_{12} = I_{12} \cdot f_{11} + I_{13} \cdot f_{12} + I_{14} \cdot f_{13} + \dots + I_{34} \cdot f_{33}$$

Applying a linear filter



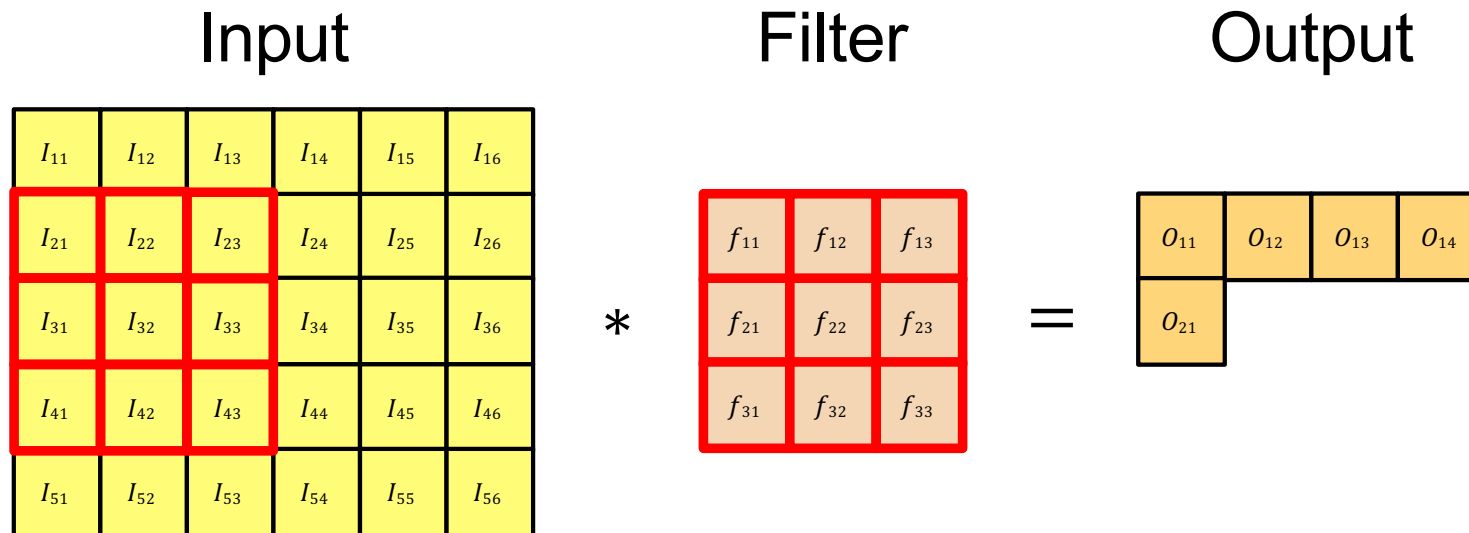
$$O_{13} = I_{13} \cdot f_{11} + I_{14} \cdot f_{12} + I_{15} \cdot f_{13} + \dots + I_{35} \cdot f_{33}$$

Applying a linear filter



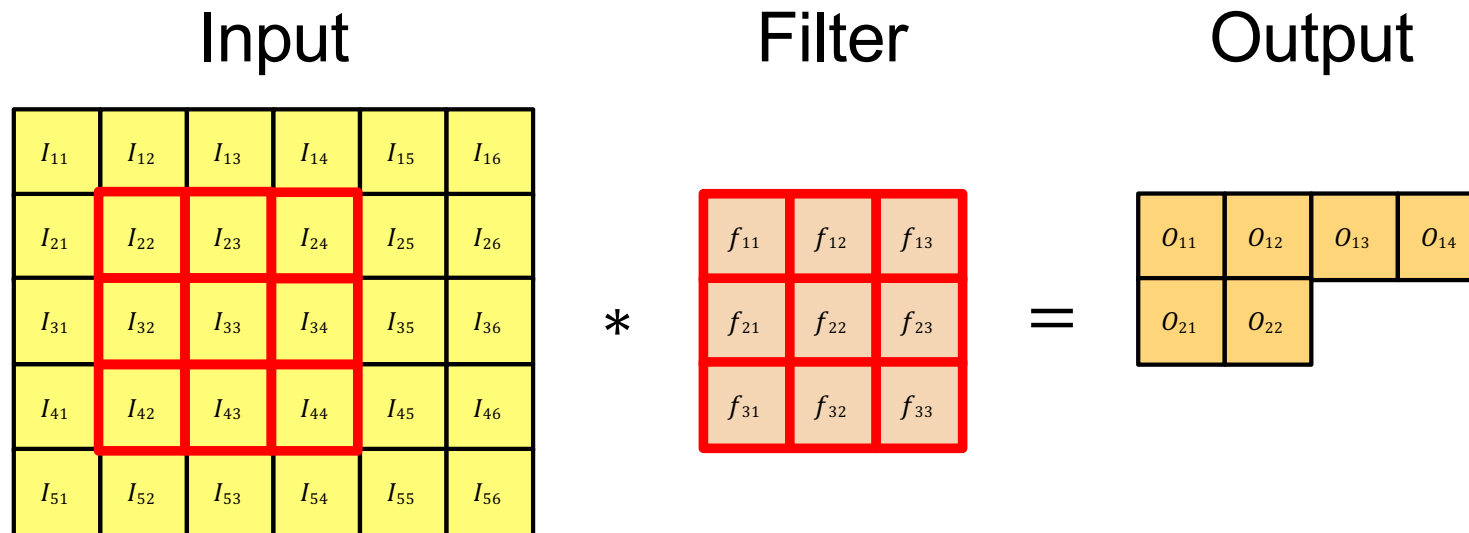
$$O_{14} = I_{14} \cdot f_{11} + I_{15} \cdot f_{12} + I_{16} \cdot f_{13} + \dots + I_{36} \cdot f_{33}$$

Applying a linear filter



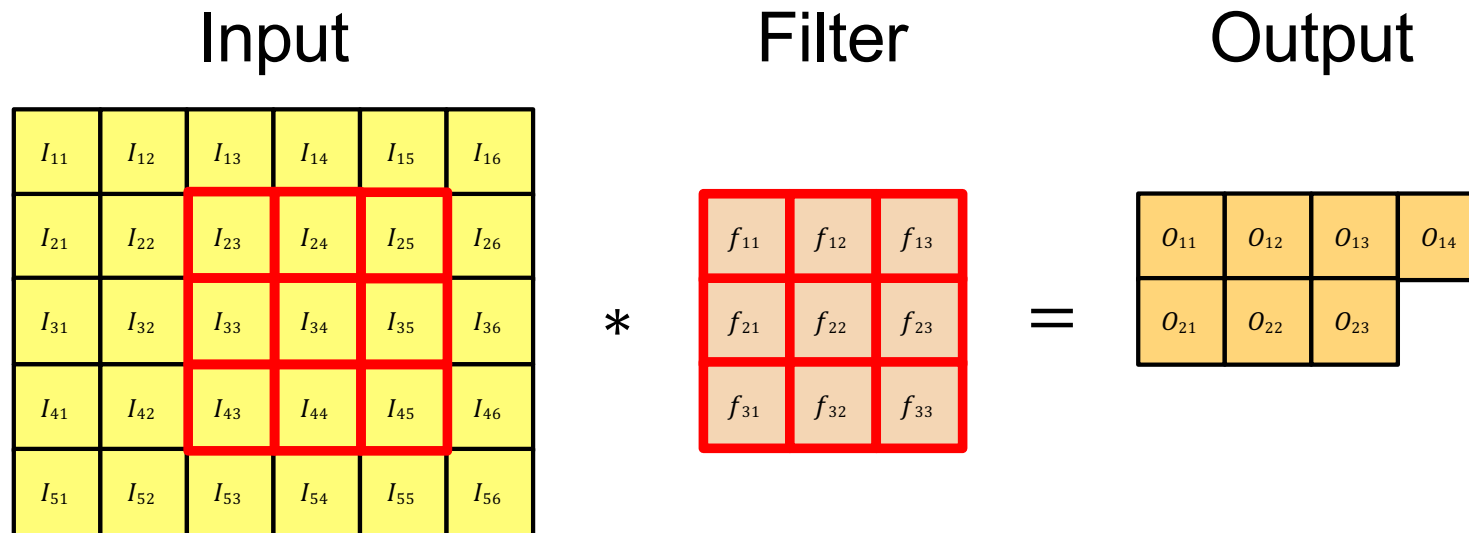
$$O_{21} = I_{21} \cdot f_{11} + I_{22} \cdot f_{12} + I_{23} \cdot f_{13} + \dots + I_{43} \cdot f_{33}$$

Applying a linear filter



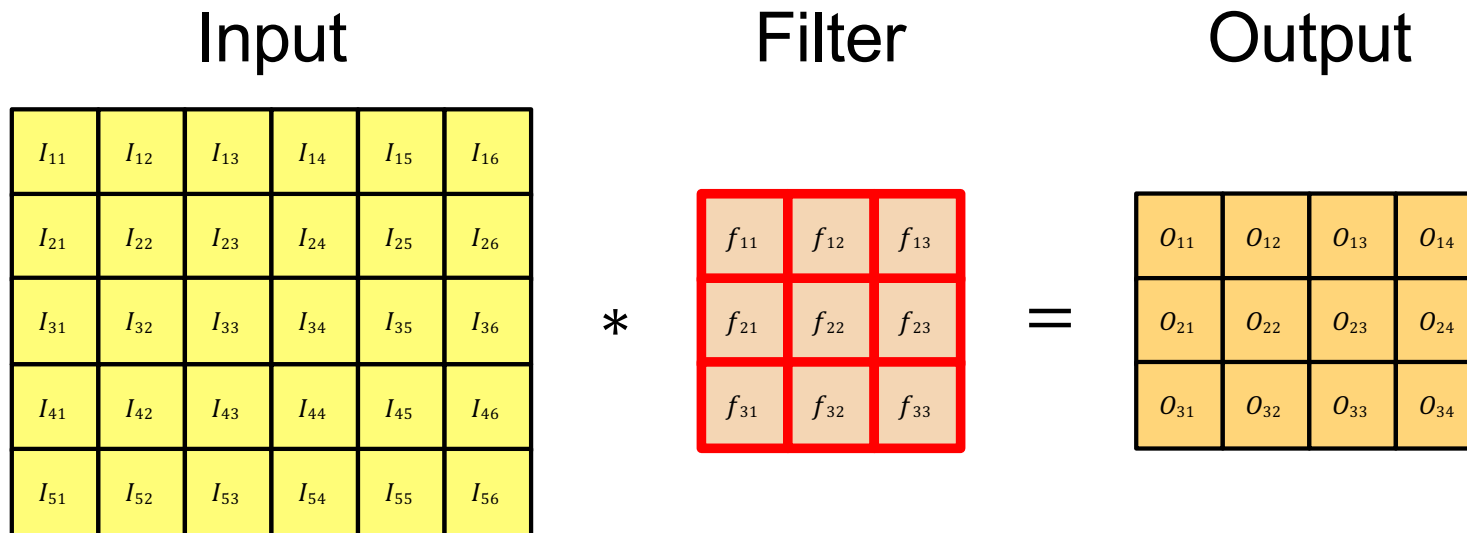
$$O_{22} = I_{22} \cdot f_{11} + I_{23} \cdot f_{12} + I_{24} \cdot f_{13} + \dots + I_{44} \cdot f_{33}$$

Applying a linear filter



$$O_{23} = I_{23} \cdot f_{11} + I_{24} \cdot f_{12} + I_{25} \cdot f_{13} + \dots + I_{45} \cdot f_{33}$$

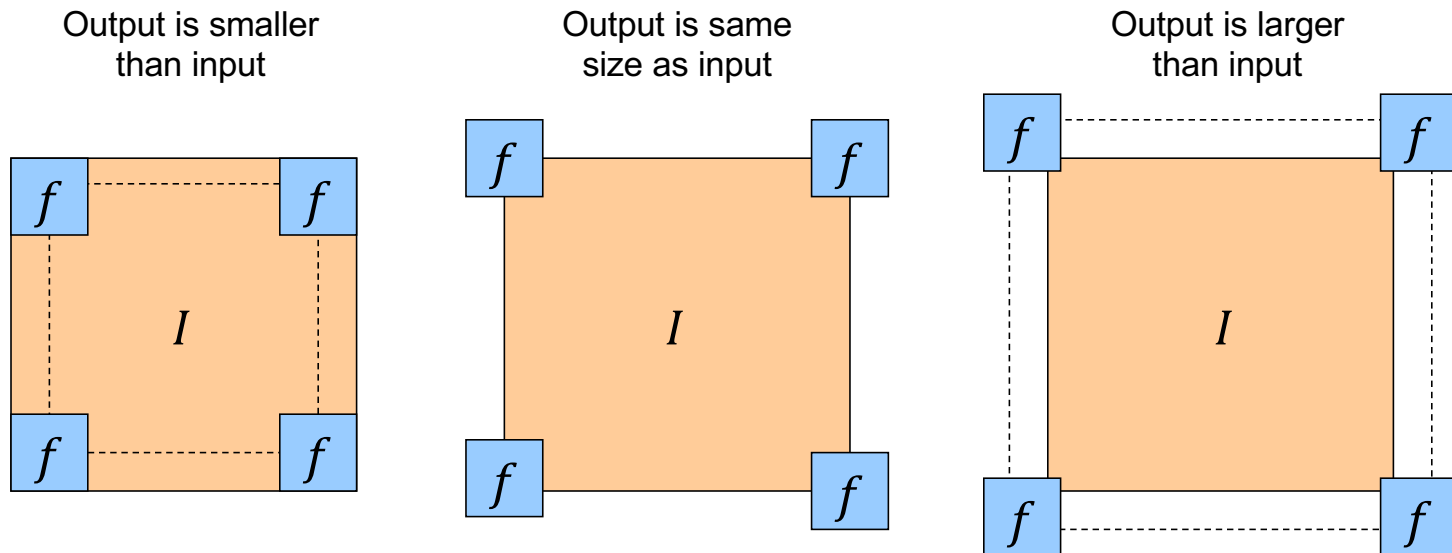
Applying a linear filter



What filter values should we use to find the average in a 3×3 window?

Practical details: Dealing with edges

- To control the size of the output, we need to use *padding*

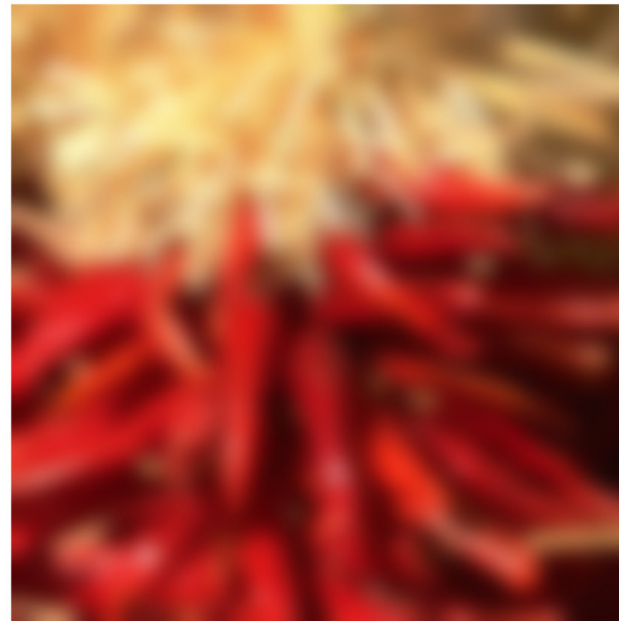


Practical details: Dealing with edges

- To control the size of the output, we need to use *padding*
- What values should we pad the image with?

Practical details: Dealing with edges

- To control the size of the output, we need to use *padding*
- What values should we pad the image with?
 - Zero pad (or clip filter)
 - Wrap around
 - Copy edge
 - Reflect across edge



Source: S. Marschner

Properties: Linearity

$$\text{filter}(I, f_1 + f_2) = \text{filter}(I, f_1) + \text{filter}(I, f_2)$$

$$\text{filter}\left(\begin{array}{|c|} \hline \blacksquare \\ \hline \end{array}, \begin{array}{|c|} \hline \blacksquare \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \end{array}\right) = \text{filter}\left(\begin{array}{|c|} \hline \blacksquare \\ \hline \end{array}, \begin{array}{|c|} \hline \square \\ \hline \end{array}\right) = \begin{array}{|c|} \hline \text{blurred square} \\ \hline \end{array}$$

$$\text{filter}\left(\begin{array}{|c|} \hline \blacksquare \\ \hline \end{array}, \begin{array}{|c|} \hline \blacksquare \\ \hline \end{array}\right) + \text{filter}\left(\begin{array}{|c|} \hline \blacksquare \\ \hline \end{array}, \begin{array}{|c|} \hline \square \\ \hline \end{array}\right) = \begin{array}{|c|} \hline \text{blurred square} \\ \hline \end{array} + \begin{array}{|c|} \hline \text{blurred square} \\ \hline \end{array} = \begin{array}{|c|} \hline \text{blurred square} \\ \hline \end{array}$$

Properties: Linearity

$$\text{filter}(I, f_1 + f_2) = \text{filter}(I, f_1) + \text{filter}(I, f_2)$$

Also:

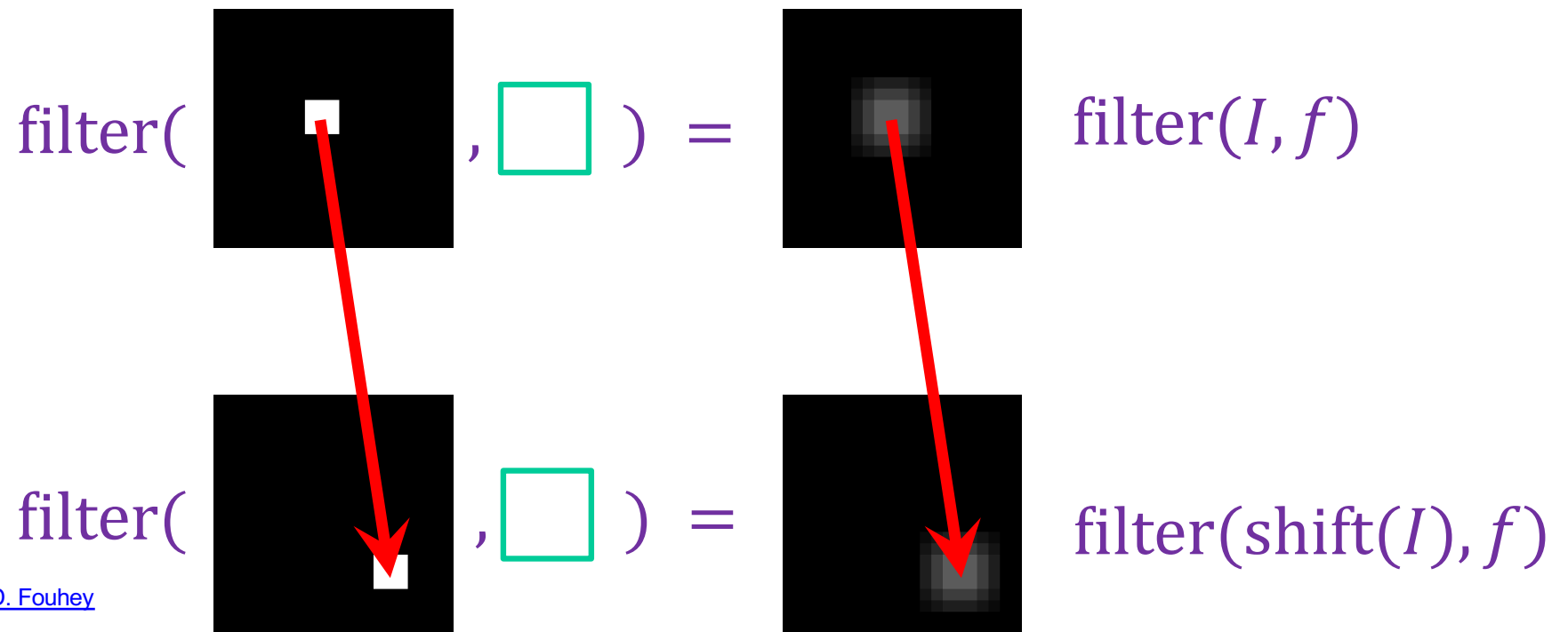
$$\text{filter}(I_1 + I_2, f) = \text{filter}(I_1, f) + \text{filter}(I_2, f)$$

$$\text{filter}(kI, f) = k \text{ filter}(I, f)$$

$$\text{filter}(I, kf) = k \text{ filter}(I, f)$$

Properties: Shift-invariance

$$\text{filter}(\text{shift}(I), f) = \text{shift}(\text{filter}(I, f))$$



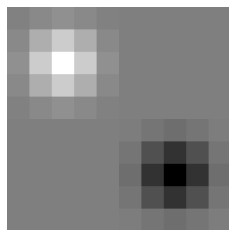
More linear filtering properties

- Commutativity: $f * g = g * f$
 - For infinite signals, no difference between filter and signal
- Associativity: $f * (g * h) = (f * g) * h$
 - Convolving several filters one after another is equivalent to convolving with one combined filter:
$$\left((g * f_1) * f_2 \right) * f_3 = g * (f_1 * f_2 * f_3)$$
- Identity: for *unit impulse* e , $f * e = f$

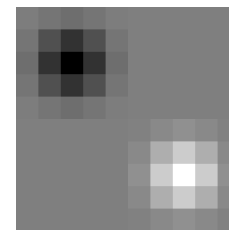
Note: Filtering vs. “convolution”

- In classical signal processing terminology, convolution is filtering with a *flipped* kernel, and filtering with an upright kernel is known as *cross-correlation*
 - Check convention of filtering function you plan to use!

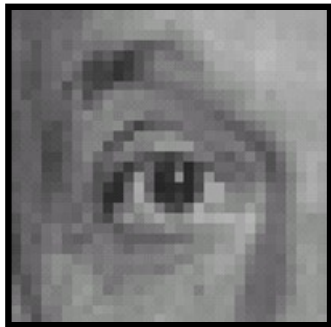
Filtering or “cross-correlation”
(Kernel in original orientation)



“Convolution”
(Kernel flipped in x and y)



Practice with linear filters

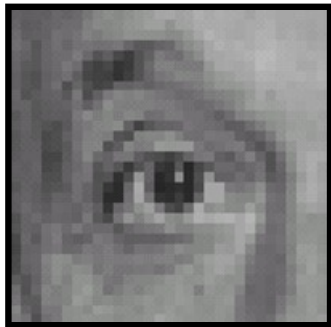


Original

0	0	0
0	1	0
0	0	0

?

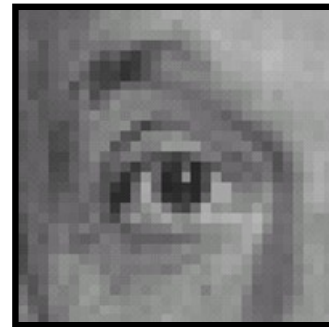
Practice with linear filters



Original

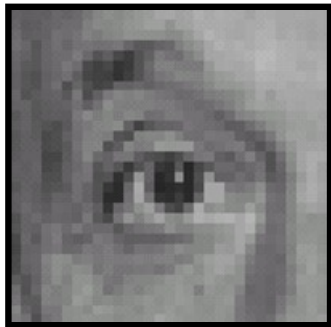
0	0	0
0	1	0
0	0	0

One surrounded
by zeros is the
identity filter



Filtered
(no change)

Practice with linear filters

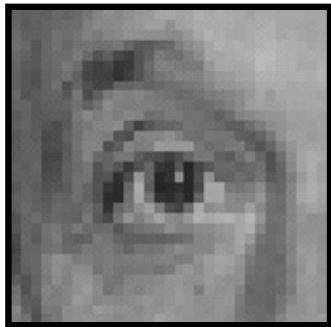


Original

0	0	0
0	0	1
0	0	0

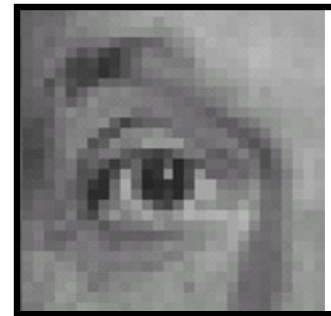
?

Practice with linear filters



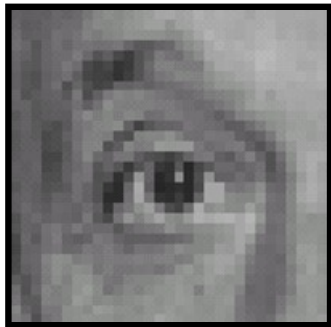
Original

0	0	0
0	0	1
0	0	0



Shifted *left*
By one pixel

Practice with linear filters



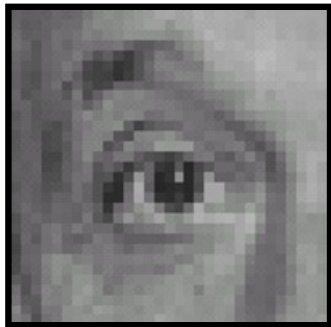
Original

 $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

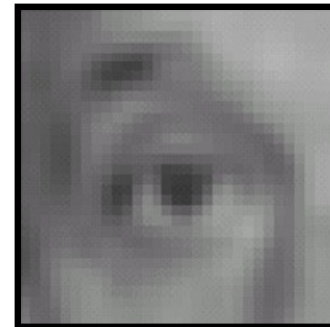
Practice with linear filters



Original

 $\frac{1}{9}$

1	1	1
1	1	1
1	1	1



Blur (with a
box filter)

Image filtering: Outline

- Linear filtering and its properties
- Gaussian filters and their properties
- Nonlinear filtering: Median filtering, bilateral filtering
- Fun filtering application: Hybrid images

Practice with linear filters



Original

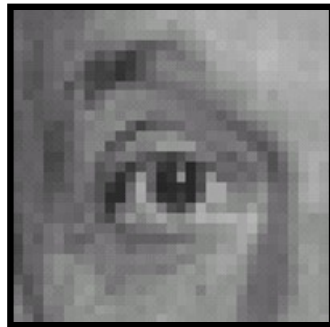
0	0	0
0	2	0
0	0	0

— $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

?

Practice with linear filters



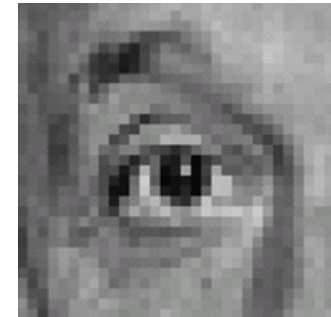
Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Sharpening filter:

Accentuates differences
with local average

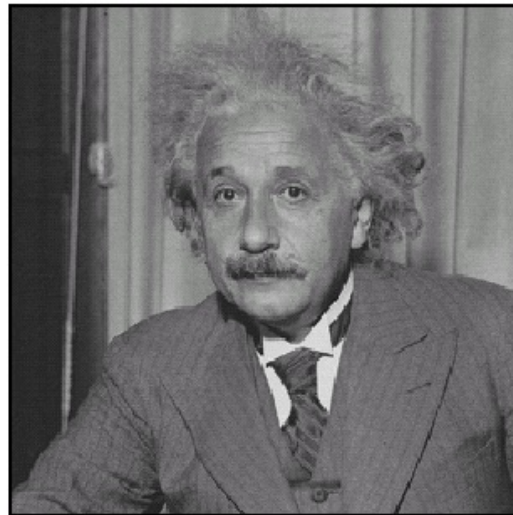
(Note that filter sums to 1)



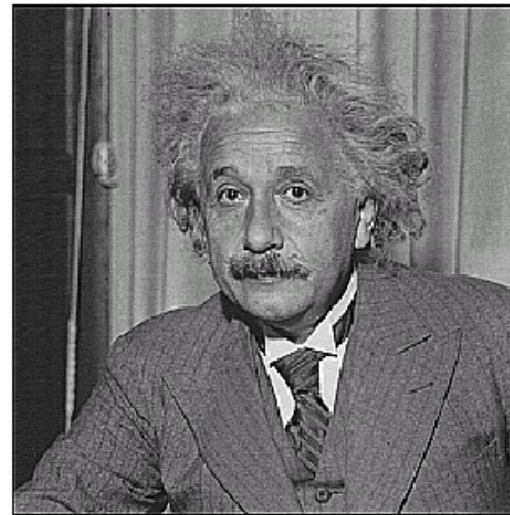
Sharpened

Sharpening

Original



Sharpened



Source: D. Lowe

Sharpening in more detail

Original

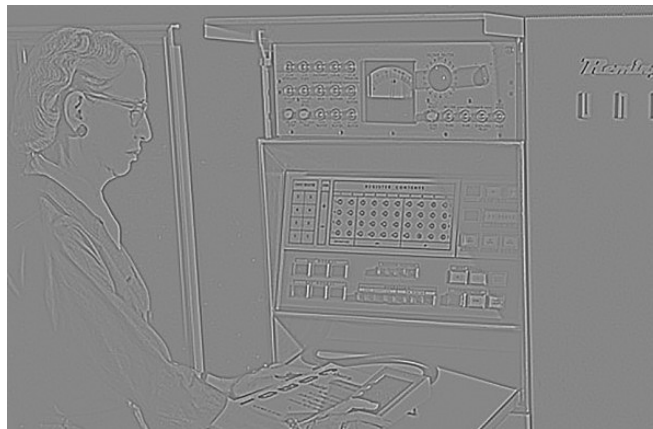


Smoothed



Details

=



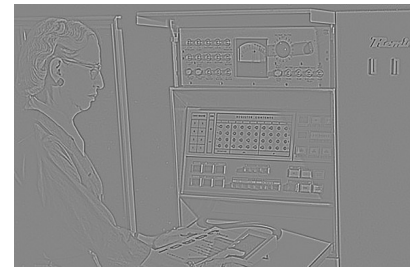
Source: D. Fouhey

Sharpening in more detail

Original



Details



$+ \alpha$

Sharpened ($\alpha = 1$)

=



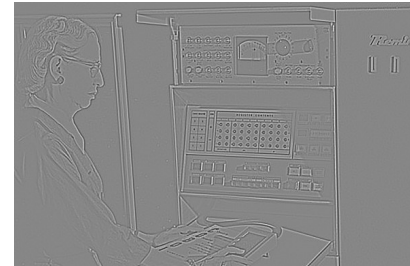
Source: D. Fouhey

Sharpening in more detail

Original



Details



$+ \alpha$

Sharpened ($\alpha = 2$)

=



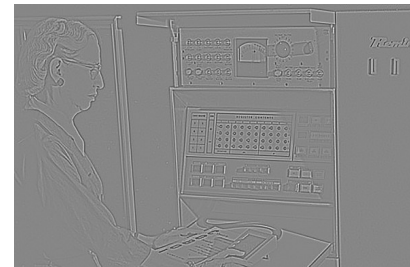
Source: D. Fouhey

Sharpening in more detail

Original



Details



$+ \alpha$

Sharpened ($\alpha = 10$)

=



Source: D. Fouhey

Image filtering: Outline

- Linear filtering and its properties
- Gaussian filters and their properties

Smoothing revisited

- What's wrong with this picture?
- What's the solution?



Source: D. Forsyth

Smoothing revisited

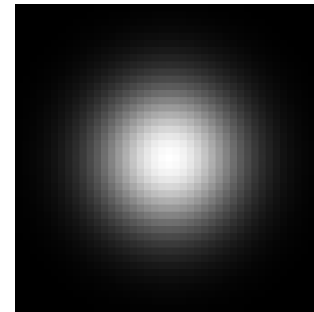
- What's wrong with this picture?
- What's the solution?
 - To eliminate box artifacts, weight contribution of neighborhood pixels according to their closeness to the center

“proportional to”
(renormalize values to sum to 1)

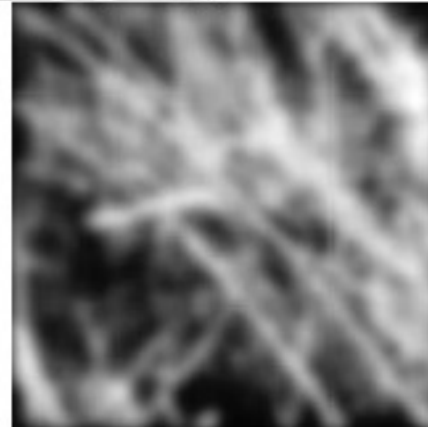
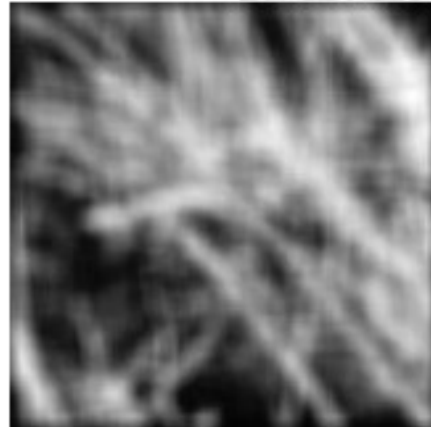
$$G(x, y) \propto \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

standard deviation
(determines size of “blob”)

Gaussian filter

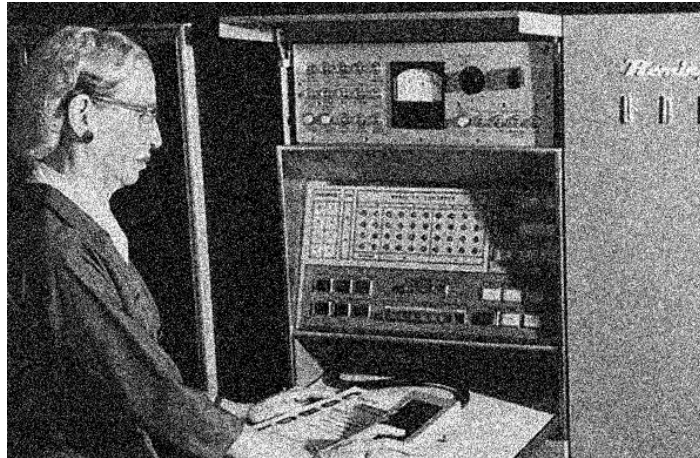


Gaussian vs. box filtering



Applying Gaussian filters

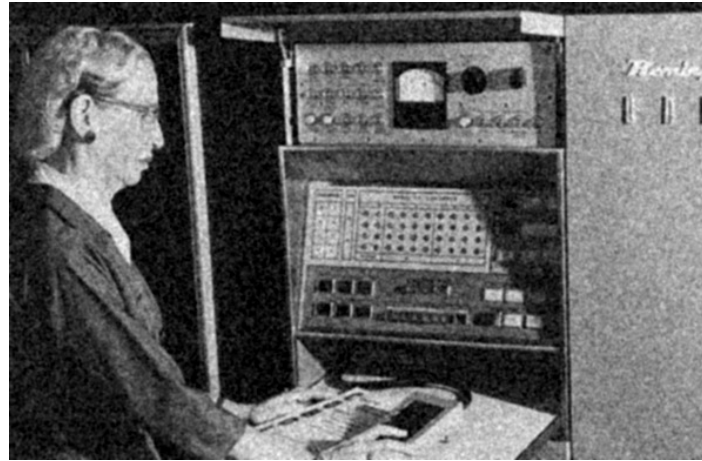
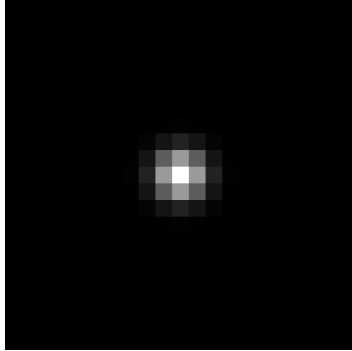
Input image
(no filter)



Source: [D. Fouhey and J. Johnson](#)

Applying Gaussian filters

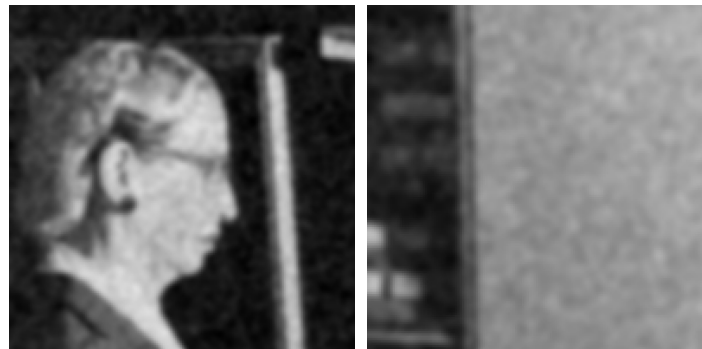
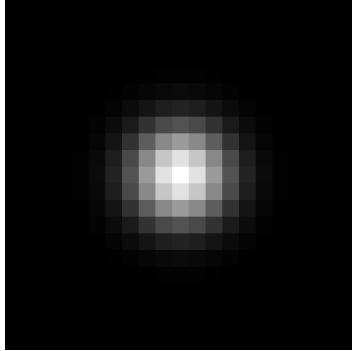
$$\sigma = 1$$



Source: [D. Fouhey and J. Johnson](#)

Applying Gaussian filters

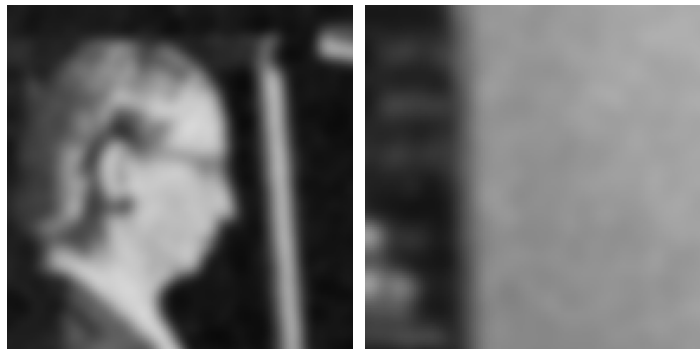
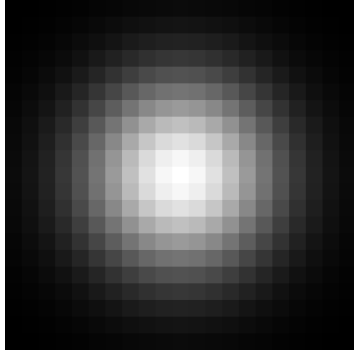
$$\sigma = 2$$



Source: [D. Fouhey and J. Johnson](#)

Applying Gaussian filters

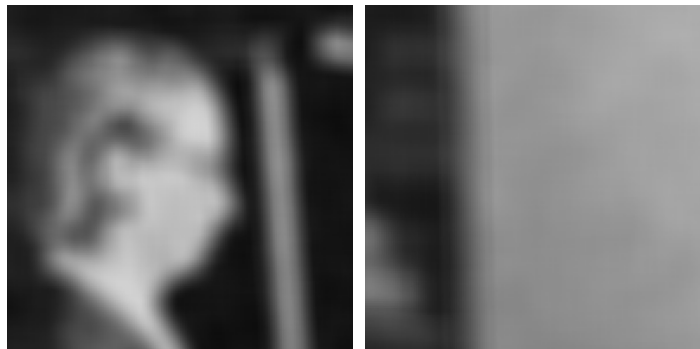
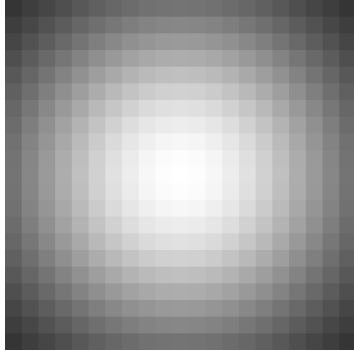
$$\sigma = 4$$



Source: [D. Fouhey and J. Johnson](#)

Applying Gaussian filters

$$\sigma = 8$$

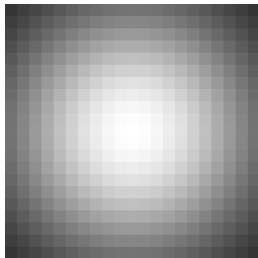


Source: [D. Fouhey and J. Johnson](#)

Choosing filter size

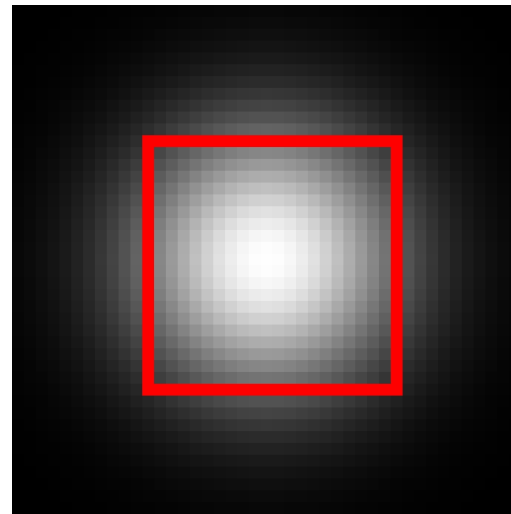
- Rule of thumb: set filter width to about 6σ (captures 99.7% of the energy)

$\sigma = 8$
Width = 21



Too small!

$\sigma = 8$
Width = 43



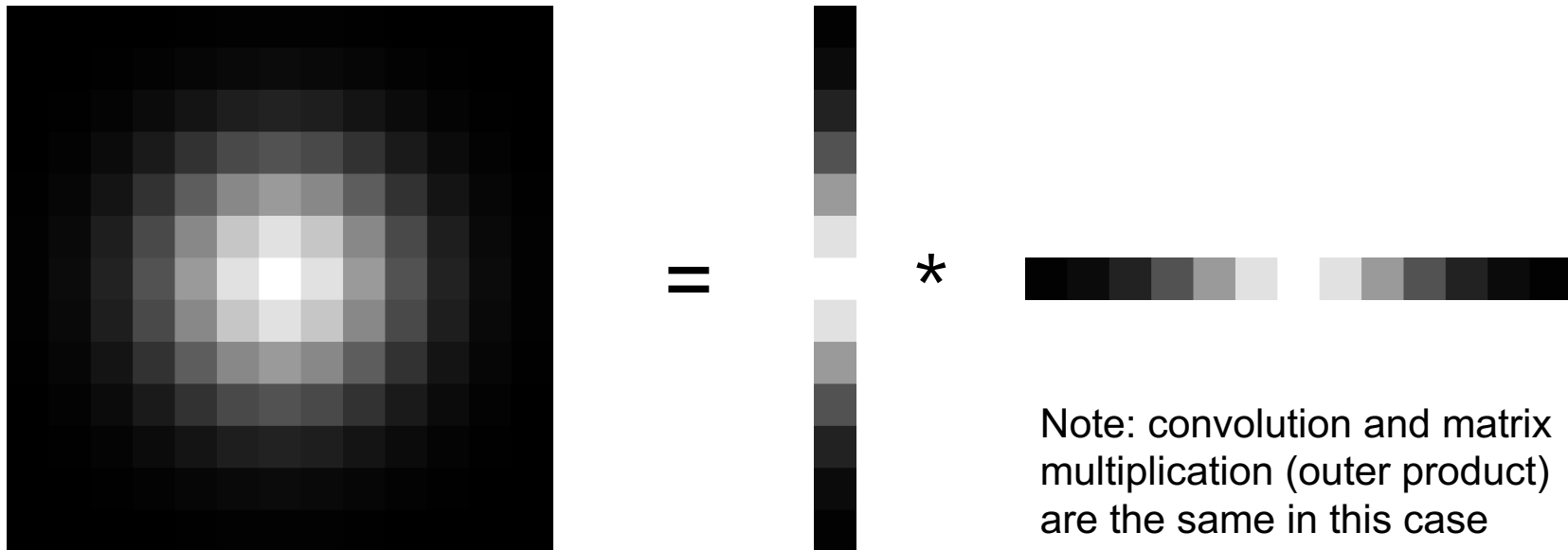
A bit small (might be OK)

Gaussian filters: Properties

- Gaussian is a *low-pass filter*: it removes high-frequency components from the image (more on this soon)
- Convolution with self is another Gaussian
 - So we can smooth with small- σ kernel, repeat, and get same result as larger- σ kernel would have
 - Convolving *two times* with Gaussian kernel with std. dev. σ is the same as convolving *once* with kernel with std. dev. $\sigma\sqrt{2}$
- Gaussian kernel is *separable*: it factors into product of two 1D Gaussians

Separability of the Gaussian filter

$$\frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2}{2\sigma^2}\right)$$



Note: convolution and matrix multiplication (outer product) are the same in this case

Why is separability useful?

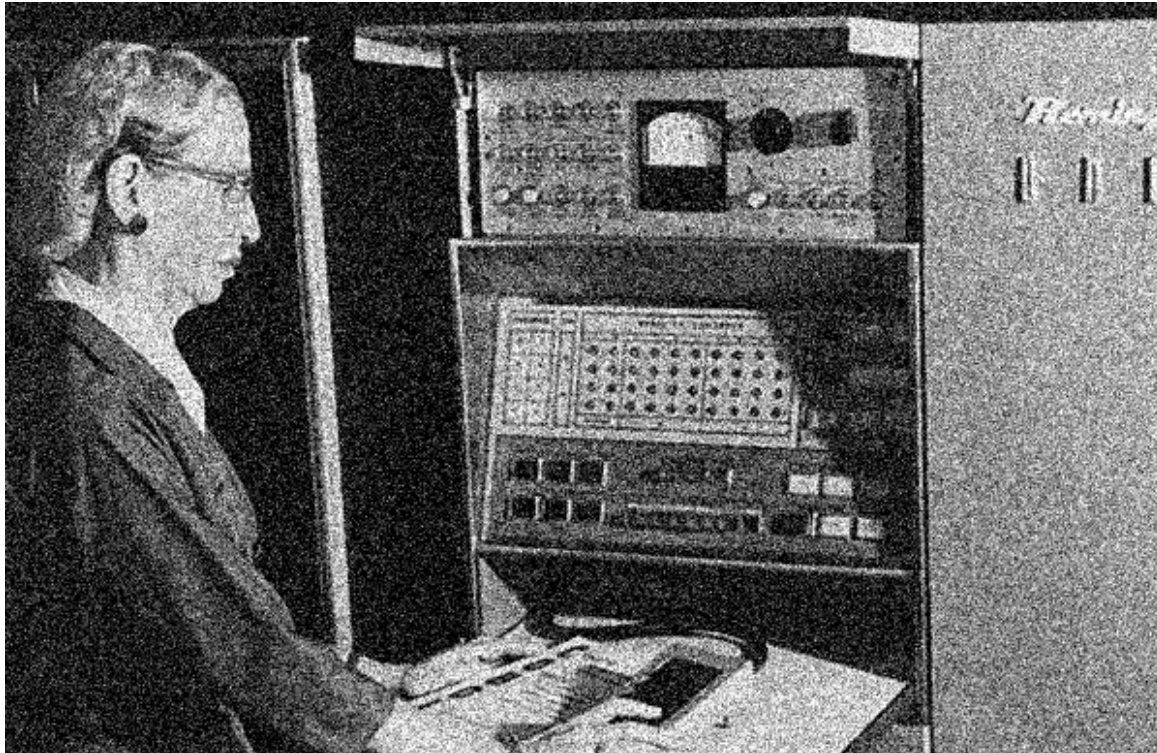
- Separability means that a 2D convolution can be reduced to two 1D convolutions (one along rows and one along columns)
- What is the complexity of filtering an $n \times n$ image with an $m \times m$ kernel?
 - $O(n^2 m^2)$
- What if the kernel is separable?
 - $O(n^2 m)$

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Different types of noise

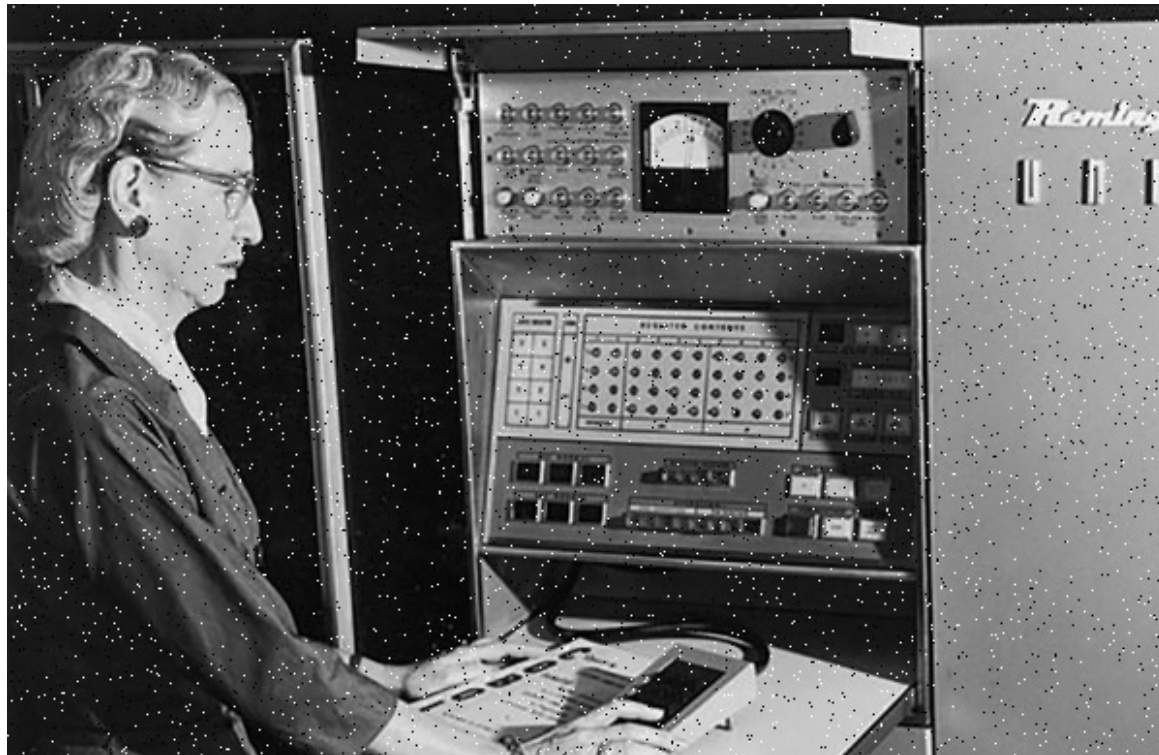
- What kind of noise is Gaussian filtering appropriate for?
 - *Additive, zero-mean* noise (assuming nearby pixels share the same value)



Adapted from [D. Fouhey and J. Johnson](#)

Different types of noise

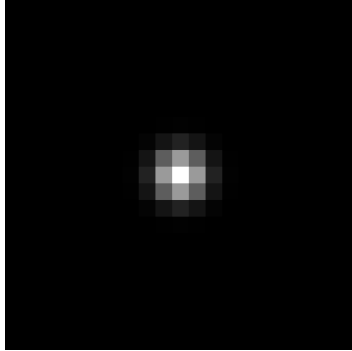
- What about *impulse* or *shot noise*, i.e., when some pixels are arbitrarily replaced by spurious values?



Adapted from [D. Fouhey and J. Johnson](#)

Where Gaussian filtering fails

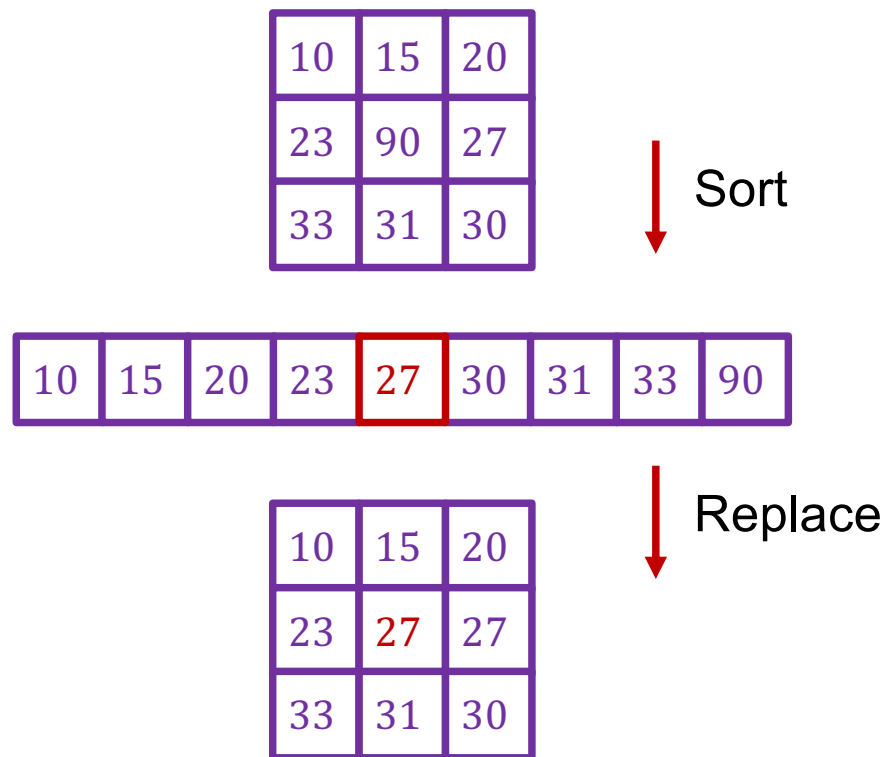
$$\sigma = 1$$



Adapted from [D. Fouhey and J. Johnson](#)

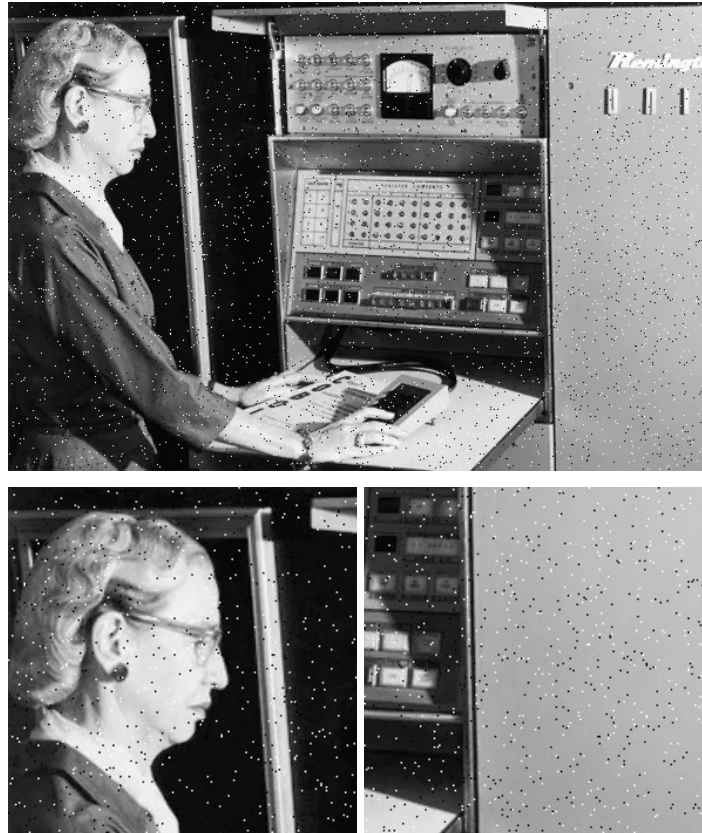
Alternative idea: Median filtering

- A **median filter** operates over a window by selecting the median intensity in the window



Applying median filter

Input image
(no filter)



Adapted from [D. Fouhey and J. Johnson](#)

Applying median filter

median
filter
(width = 3)



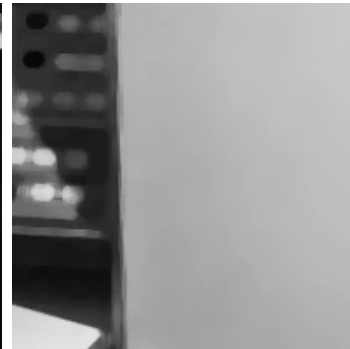
Applying median filter

median
filter
(width = 5)



Applying median filter

median
filter
(width = 7)



Is median filtering linear?

- Recall: for a linear filter,

$$\text{filter}(I_1 + I_2) = \text{filter}(I_1) + \text{filter}(I_2)$$

Is median filtering linear?

- Recall: for a linear filter,

$$\text{filter}(I_1 + I_2) = \text{filter}(I_1) + \text{filter}(I_2)$$

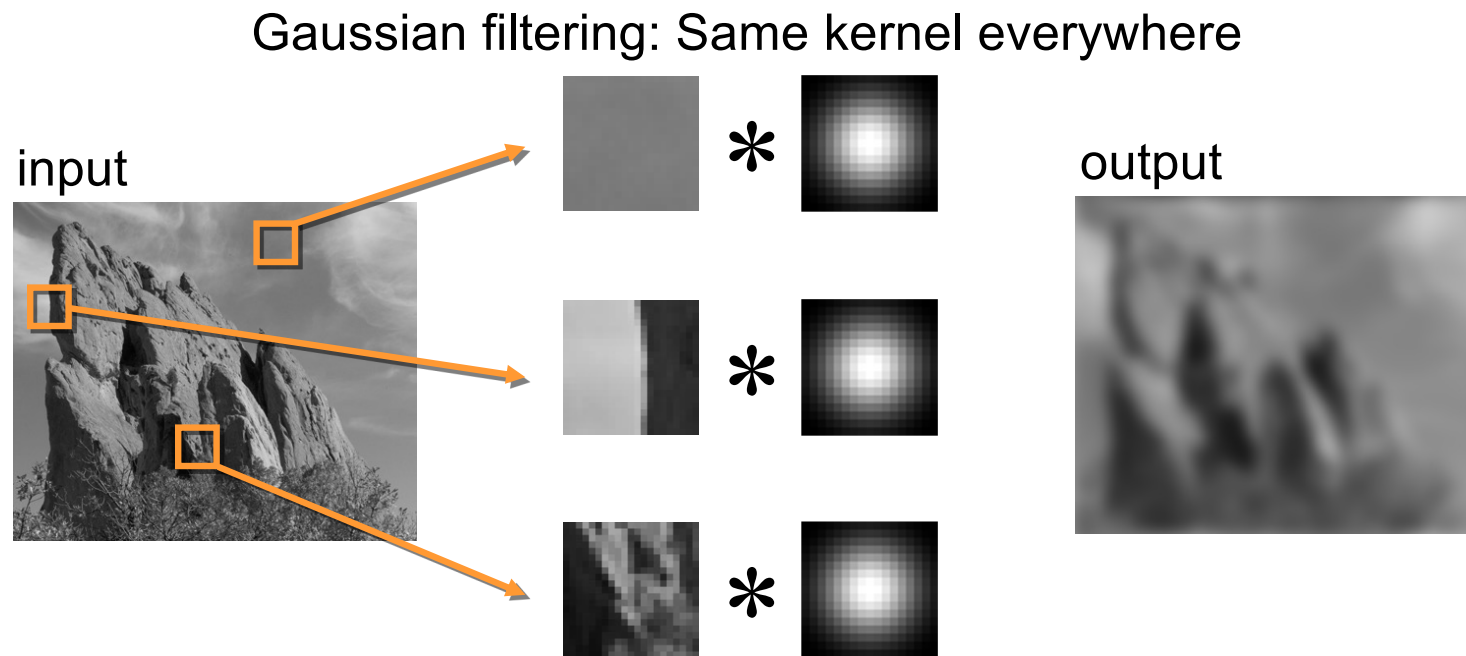
$$\begin{array}{ccccc} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 2 & 2 \end{bmatrix} & + & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} & = & \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \\ \text{median} & & & & \\ \text{filter} & & & & \\ \downarrow & & \downarrow & & \downarrow \\ 1 & + & 0 & \neq & 2 \end{array}$$

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- Nonlinear filtering: Median filtering, bilateral filtering

What other kinds of filtering are there?

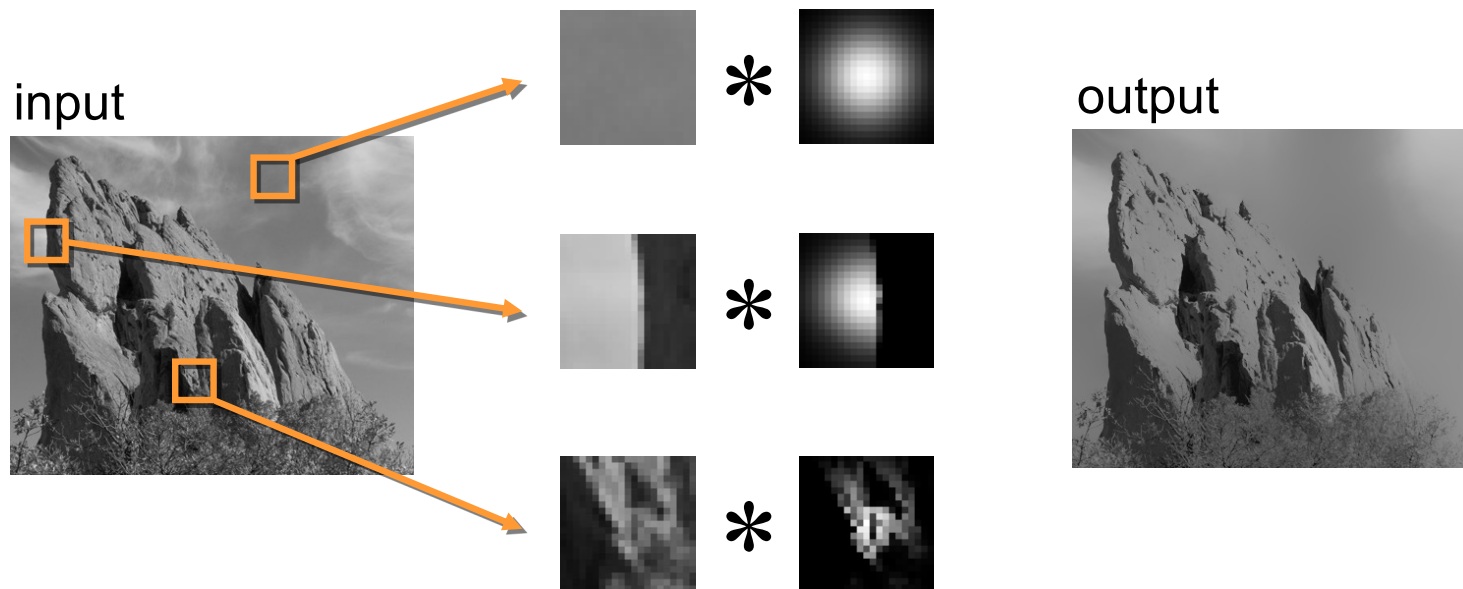
- We want something like Gaussian filtering, but with edge preservation



Bilateral filtering

[Aurich 95, Smith 97, Tomasi 98]

Kernel weights depend on the neighborhood values – no averaging across edges!



Bilateral filtering

- Compute weighted average of pixels, but the weight depends both on the distance of the filter element from the center and the difference of pixel value from the center

$$BF[I]_p = \frac{1}{W_p} \sum_{q \in S} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}(|I_p - I_q|) I_q$$

Diagram illustrating the components of the Bilateral Filtering equation:

- normalization factor**: $\frac{1}{W_p}$
- space weight**: $G_{\sigma_s}(\|p - q\|)$ (represented by a Gaussian image)
- range weight**: $G_{\sigma_r}(|I_p - I_q|)$ (represented by a graph of a Gaussian function over intensity I)

Bilateral filtering

input



$\sigma_s = 2$



$\sigma_r = 0.1$



$\sigma_r = 0.25$

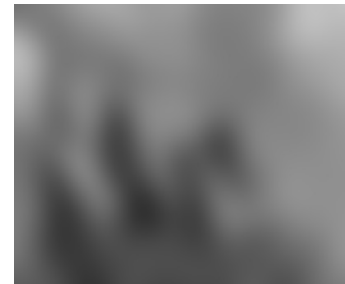
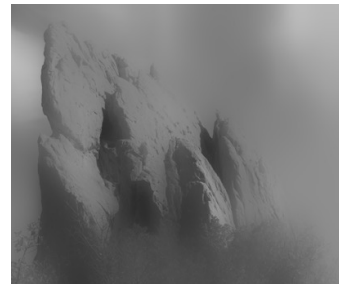
$\sigma_r = \infty$ (Gaussian blur)



$\sigma_s = 6$



$\sigma_s = 18$



Source: [SIGGRAPH 2008 tutorial](#)

Denoising comparison

Noisy input



Median 3x3



Median 5x5



Bilateral 7x7



Source: [SIGGRAPH 2008 tutorial](#)

Image filtering: Outline

- Linear filtering and its properties
- Gaussian filters and their properties
- Nonlinear filtering: Median filtering, bilateral filtering
- Fun filtering application: Hybrid images

Application: Hybrid images



A. Oliva et al. [Hybrid Images](#). SIGGRAPH 2006

Recall: Sharpening

Image



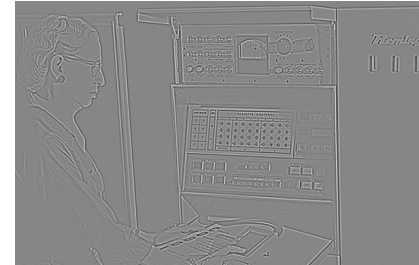
Smoothed



-

=

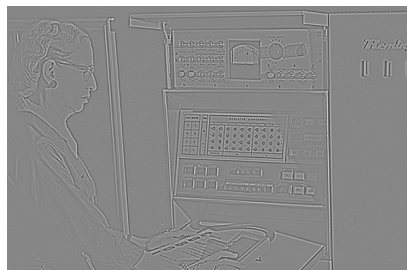
Details



Image



Details



+ α

=

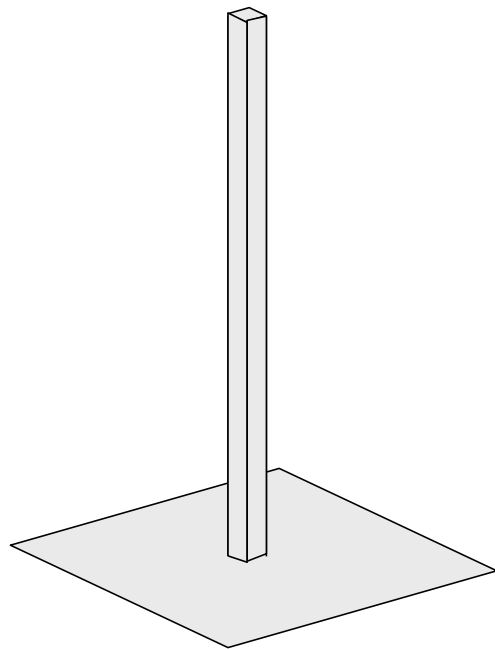
Sharpened



“Detail” filter

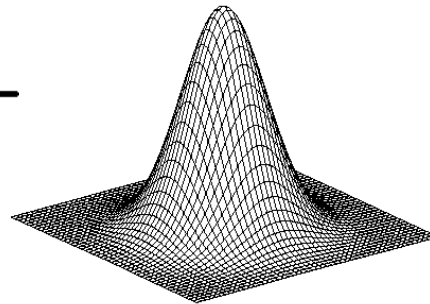
$$I - I * g = I * (e - g)$$

↑
unit impulse
(identity)



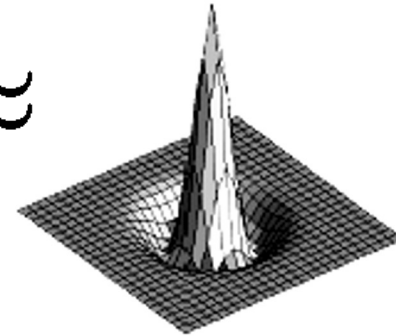
unit impulse

—



Gaussian

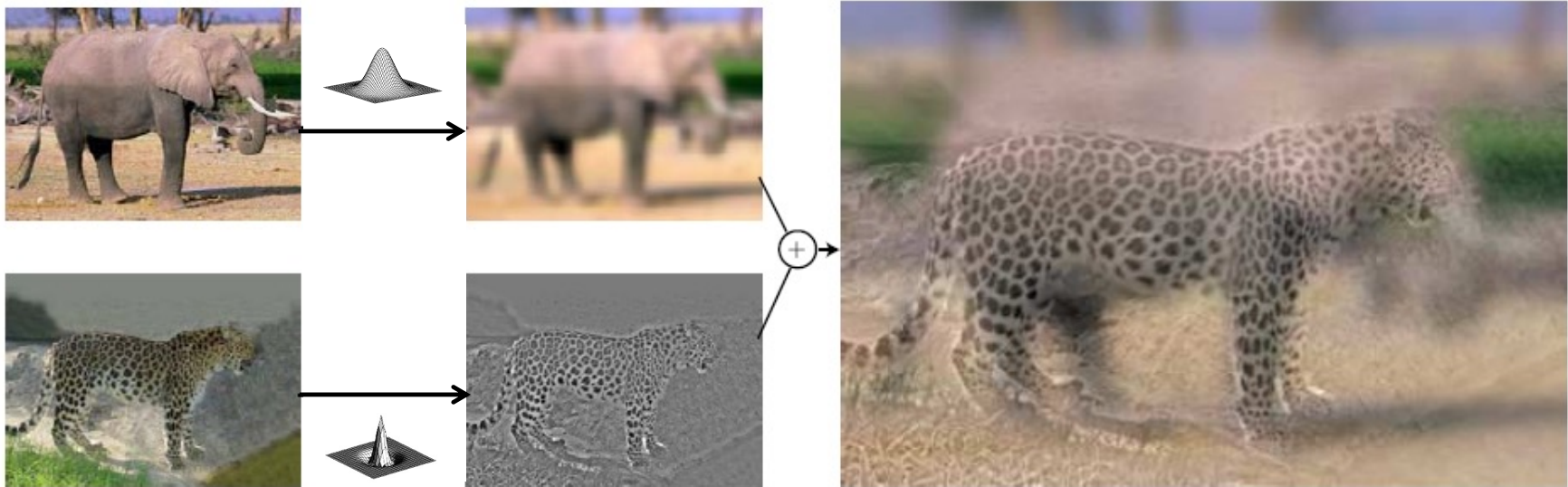
≈



Laplacian of Gaussian

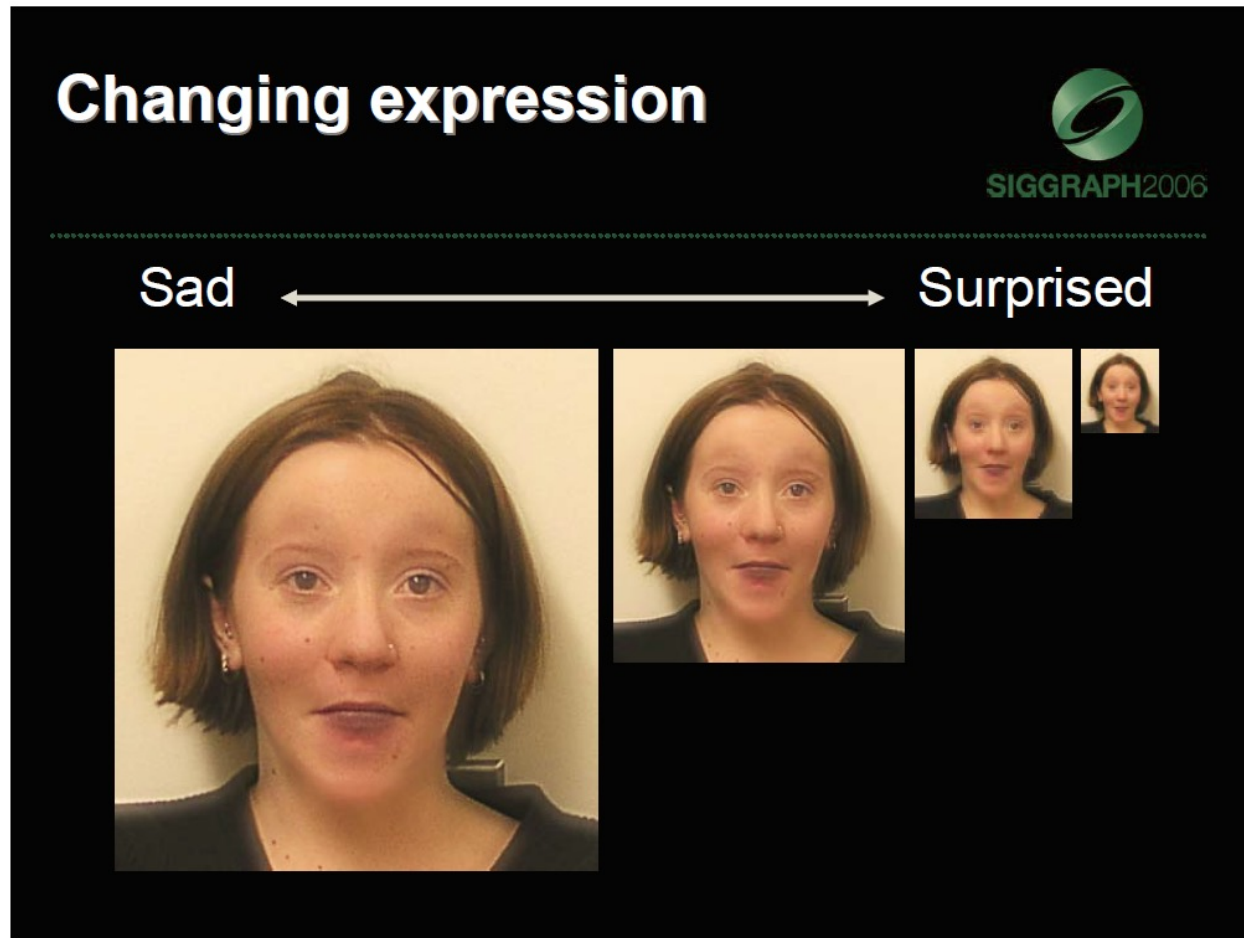
Application: Hybrid images

Gaussian filter



Laplacian filter

Application: Hybrid images



Why it's good to read old papers



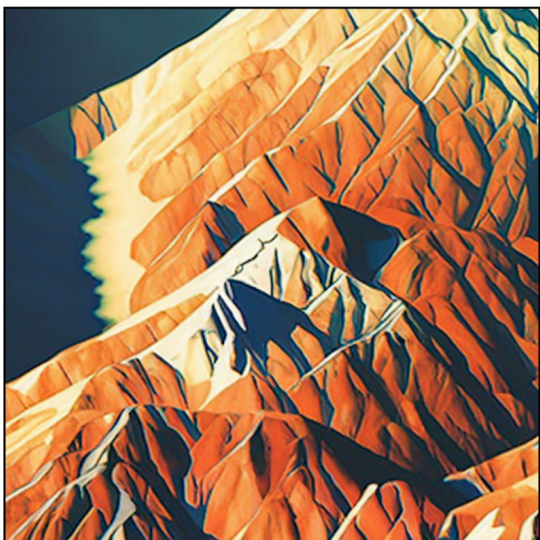
© Quint Buchholtz



Alexei (Alyosha) Efros
UC Berkeley

[RetroCV workshop at CVPR 2024](#)

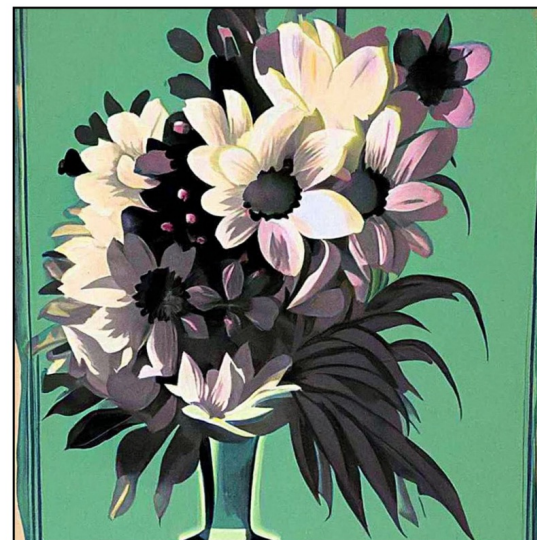
Hybrid images with diffusion models



"a photo of mountains"



"a lithograph of
waterfalls"



"a lithograph of
flower arrangements"



"a photo of a
teddy bear"



"a lithograph
of a pig"



"a lithograph
of a panda"

Hybrid images with diffusion models



"a photo of
houseplants"



"a photo of a rabbit"



"a photo of
new york city"



"a photo of
marilyn
monroe"



"a photo of
an old man"



"a photo of
john lennon"

D. Geng et al. [Factorized Diffusion: Perceptual Illusions by Noise Decomposition](#). ECCV 2024

Hybrid images with diffusion models

Triple hybrids

"a photo of
waterfalls"

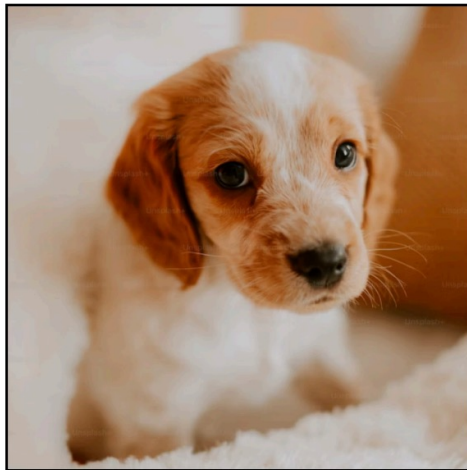


D. Geng et al. [Factorized Diffusion: Perceptual Illusions by Noise Decomposition](#). ECCV 2024

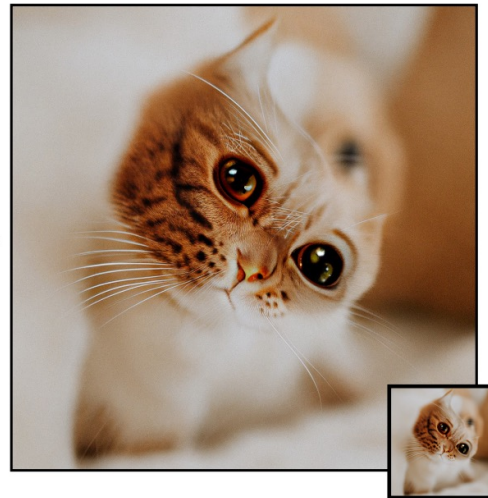
Hybrid images with diffusion models

Inverse hybrids

Low Pass:
[Real Image]



High Pass:
"a photo of a cat"



D. Geng et al. [Factorized Diffusion: Perceptual Illusions by Noise Decomposition](#). ECCV 2024

Hybrid images with diffusion models

Comparison with Oliva et al. (2006)

“leopard” / “elephant”



Geng et al. (2024)



Oliva et al. (2006)

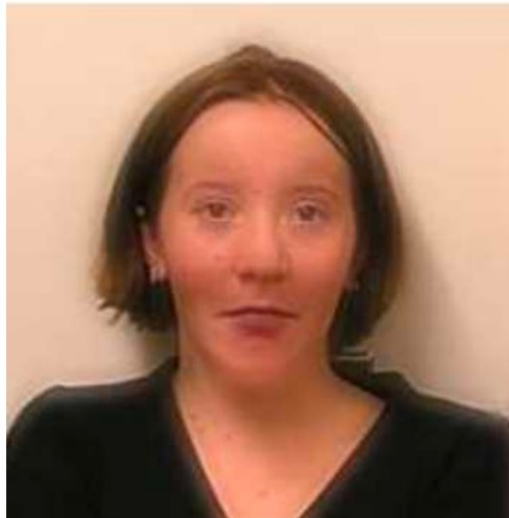
Hybrid images with diffusion models

Comparison with Oliva et al. (2006)

“sad woman” / “happy woman”



Geng et al. (2024)

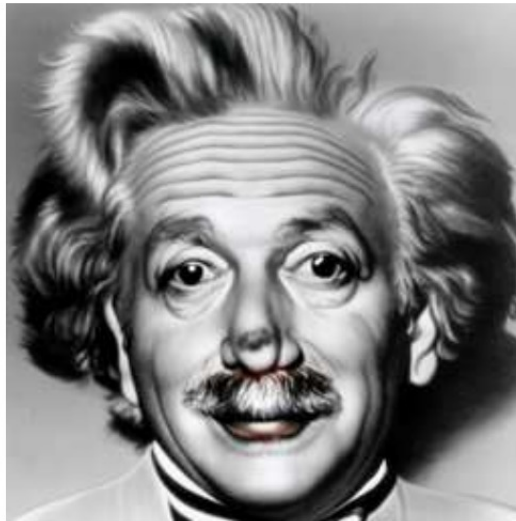


Oliva et al. (2006)

Hybrid images with diffusion models

Comparison with Oliva et al. (2006)

“Albert Einstein” / “Marilyn Monroe”



Geng et al. (2024)



Oliva et al. (2006)

Hybrid images with diffusion models

Factorized diffusion

